

Algebra (8 th Std)

Choose the Appropriate Answer

- The product of $7p^3$ and $(2p^2)^2$ is
(a) $14p^{12}$ (b) $28p^7$ (c) $9p^7$ (d) $11p^{12}$
Ans: (b) $28p^7$
- The missing terms in the product $-3m^3n \times 9(_) = ______ m^4n^3$ are
(a) $mn^2, 27$ (b) $m^2n, 27$ (c) $m^2n^2, -27$ (d) $mn^2, -27$
Ans: (a) $mn^2, 27$
- If the area of a square is $36x^4y^2$ then, its side is $______$.
(a) $6x^4y^2$ (b) $8x^2y^2$ (c) $6x^2y$ (d) $-6x^2y$
Ans: (C) $6x^2y$
- If the area of a rectangle is $48m^2n^3$ and whose length is $8mn^2$ then, its breadth is $______$.
(a) $6m n$ (b) $8m^2n$ (c) $7m^2n^2$ (d) $6m^2n^2$
Ans: (a) $6m n$
- If the area of a rectangular land is $(a^2 - b^2)$ sq. Units whose breadth is $(a - b)$ then, its length is $______$
(a) $a - b$ (b) $a + b$ (c) $a^2 - b$ (d) $(a + b)^2$
Ans: (b) $a + b$
- $6xy \times ______ = -12x^3y$
(a) $2x^2$ (b) $-2x^2$ (c) $3x^2$ (d) $-3x^2$
Ans: (b) $-2x^2$
- $______ \times (-15m^2n^3p) = 45m^3n^3p^2$
(a) $-3m p$ (b) $3m p$ (c) $4m n$ (d) $2m n$
Ans: (a) $-3mp$
- If $x^2 - y^2 = 16$ and $(x + y) = 8$ then $(x - y)$ is $______$
(a) 8 (b) 3 (c) 2 (d) 1
Ans: (c) 2

$$x^2 - y^2 = 16 \Rightarrow (x + y)(x - y) = 16 \Rightarrow 8(x - y) = 16 \Rightarrow (x - y) = \frac{16}{8} = 2$$
- $(a + b)^2 = ______$
(a) $a^2 - ab + b^2$ (b) $a^2 + ab + b^2$ (c) $a^2 + 2ab + b^2$ (d) $a^2 - 2ab + b^2$
Ans: (b) $a^2 + ab + b^2$
- $(p + q)(p^2 - pq + q^2)$ is equal to $______$
(a) $p^3 + q^3$ (b) $(p + q)^3$ (c) $p^3 - q^3$ (d) $(p - q)^3$
Ans: (a) $p^3 + q^3$

11. $(a - b) = 3$ and $ab = 5$ then $a^3 - b^3 =$ _____
(a) 15 (b) 18 (c) 62 (d) 72

Ans: (d) 72

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2) = 3(19 + 5) = 3(24) = 72$$

12. $a^3 + b^3 = (a + b)^3$ _____
(a) $3a(a + b)$ (b) $3ab(a - b)$ (c) $-3ab(a + b)$ (d) $3ab(a + b)$

Ans: (d) $3ab(a + b)$

13. The value of x in the equation $x + 5 = 12$ is _____.
(a) 5 (b) 6 (c) 8 (d) 7

Ans: (d) 7

14. The value of y in the equation $y - 9 = (-5) + 7$ is _____.
(a) 9 (b) 10 (c) 11 (d) 12

Ans: (c) 11

15. The value of m in the equation $8m = 56$ is _____.
(a) 6 (b) 7 (c) 8 (d) 4

Ans: (b) 7

16. The value of p in the equation $2p^3 = 10$ is _____.
(a) 12 (b) 14 (c) 13 (d) 15

Ans: (d) 15

$$\begin{array}{l} \text{Multiplying by 3 on both sides } \frac{2p}{2} \times 3 = 10 \times 3 \\ \text{Dividing by 2 on both sides } \frac{2p}{2} = \frac{30}{2} \Rightarrow p = 15 \end{array}$$

17. The linear equation in one variable has _____ solution.
(a) 1 (b) 2 (c) 3 (d) 4

Ans: (a) 1

18. The solution of the equation $ax + b = 0$ is _____.
(a) b / a (b) a / b (c) $-b / a$ (d) $-a / b$

Ans: (c) $-b / a$

19. If 'a' and 'b' are positive integers then the solution of the equation $ax = b$ has to be always _____.
(a) Positive (b) Negative (c) Both (d) None of these

Ans: (a) Positive

20. One-sixth of a number when subtracted from the number itself gives 25. The number is _____.
(a) 25 (b) 30 (c) 35 (d) 20

Ans: (b) 30

$$\begin{aligned} \text{Let the number be } x. &\Rightarrow x - \frac{x}{6} = 25 \Rightarrow \frac{5x}{6} = 25 \\ &\Rightarrow x = \frac{25 \times 6}{5} = 5 \times 6 = 30 \end{aligned}$$

21. In an equation $a + b = 23$. The value of 'a' is 14 then the value of 'b' is _____.
 (a) 6 (b) 7 (c) 9 (d) 8

Ans: (c) 9

22. Sum of a number and its half is 30 then the number is _____.
 (a) 15 (b) 20 (c) 25 (d) 40

Ans: (b) 20

23. The largest number of the three consecutive numbers is $x + 1$, then the smallest number is _____.
 (a) x (b) $x + 1$ (c) $x + 2$ (d) $x - 1$

Answer: (d) $x - 1$

The 3 consecutive numbers are: $x - 1, x, x + 1$

24. X- axis and Y-axis intersect at _____.
 (a) Point (0,1) (b) Point (1, 0) (c) Origin (0, 0) (d) None

Ans: (c) Origin (0, 0)

25. The coordinates of the point in third quadrant are always _____.
 (a) +, - (b) +, + (c) -, - (d) -, +

Ans: (c) -, -

26. (0, -5) point lies on _____ axis.
 (a) X – axis (b) Y – axis (c) Both (d) None

Ans: (b) Y-axis

27. The x- coordinate is always _____ on the y-axis.
 (a) Straight (b) Parabola (c) Zero (d) None of these

Ans: (c) Zero

28. What do we get when we transpose $5 / 2$ to RHS in the equation $x / 4 + 5 / 2 = - 3 / 4$?

- (a) $x / 4 = -3 / 4 + 5 / 2$ (b) $x / 4 = -5 / 2 + 3 / 4$
 (c) $x / 4 = -3 / 4 + (-5 / 2)$ (d) None of these

Ans: (c) $x / 4 = -3 / 4 + (-5 / 2)$

29. In the equation $3x = 4 - x$, transposing $-x$ to LHS we get
 (a) $3x - x = 4$ (b) $3x + x = 4$ (c) $-3x + x = 4$ (d) $-3x - x = 4$

Ans: (b) $3x + x = 4$

30. If $x / 3 + 1 = 7 / 15$, then which of the following is correct?

- (a) $x / 3 = 7 / 15 - 1$ (b) $x / 3 = -7 / 15 + 1$
 (c) $x / 3 = -7 / 15 - 1$ (d) None of these

Ans: (a) $x / 3 = 7 / 15 - 1$

31. If $7x+15 = 50$, then which of the following is the root of the equation?

- (a) -5 (b) $65 / 7$ (c) 5 (d) $1 / 5$

Ans: (c) 5

32. If $2x / 5 = 4$, the value of x is

- (a) 10 (b) -10 (c) $-8 / 5$ (d) $8 / 5$

Ans: (a) 10

33. If the sum of two consecutive numbers is 71 and one number is x, then the other number is-

- (a) $x + (x+1) = 71$ (b) $x + (x+2) = 71$ (c) $x + x = 71$ (d) None of these

Ans: (a) $x + (x+1) = 71$

34. Two year ago my age was x years, then what was my age 5 years ago?

- (a) $X + 7$ (b) $X - 2 - 5$ (c) $X - 5$ (d) $X - 3$

Ans: (d) $X - 3$

35. How old will I be after 10 years, if my age before 10 years was 'x' years?

- (a) $X + 20$ (b) $X - 20$ (c) $X + 10$ (d) $X - 10$

Ans: (a) $X + 20$

36. If the difference of two consecutive number is 15 and greater of them is x then the smaller number is:

- (a) 16 (b) 14 (c) 8 (d) 7

Ans: (c) 8

37. If x is an even number, which is the next odd number?

- (a) $X + 1$ (b) $X + 2$ (c) $X - 1$ (d) $X - 2$

Ans: (a) $X + 1$

+-

38. (0, y) are the co-ordinates of a point lying on which of the following?

- (a) Origin (b) x - axis (c) y - axis (d) None of these.

Ans: (b) x - axis

39. The point (3, 2) is nearer to:

- (a) x - axis (b) y - axis (c) Origin (d) None of these

Ans: (a) x - axis

40. The point (-5, 6) is nearer it:

- (a) x - axis (b) y - axis (c) origin (d) None of these.

Ans: (b) y - axis

41. The point (-3, -3) is

- (a) Nearer to x - axis (b) y - axis
 (c) Near to origin (d) Equidistant from x-axis and y-axis

Ans: (d) Equidistant from x-axis and y-axis

42. The point (0, 4) lies on which of the following:

- (a) x - axis (b) y - axis (c) Origin (d) None of these

Ans: (b) y - axis

43. The point (-3, 0) lies on which of the following?

- (a) x-axis (b) y - axis (c) Origin (d) None of these

Ans: (a) x - axis

44. The points (-3, 2) and (2, -3) represent:

- (a) Different points (b) Same point (c) Origin (d) None of these

Ans: (a) Different points

45. By joining (-1, -1), (0, 0) and (3, 3) represent:

- (a) A triangle (b) A straight line passing through origin
(d) A straight line not passing through origin. (d) A curved line

Ans: (b) A straight line passing through origin

46. By joining (-3, 2), (-3, -3) and (-3, 4), which of the following is obtained?

- (a) A triangle (b) A straight line not passing through origin
(c) A straight line passing through origin (d) None of these.

Ans: (b) A straight line not passing through origin

47. Which of the following points lies on y-axis?

- (a) (-4, 0) (b) (4, 0) (c) (0, -4) (d) (-4, 4)

Ans: (c) (0, -4)

48. Which of the following is the numerical coefficient of x^2y^2 ?

- (a) 0 (b) 1 (c) x^2 (d) y^2

Ans: (b) 1

49. Which of the following is the numerical coefficient of $-5xy$?

- (a) 5 (b) $-x$ (c) -5 (d) $-y$

Ans: (c) -5

50. $p^2q + r$ is what type of polynomial?

- (a) Monomial (b) Binomial (c) Trinomial (d) None of these

Ans: (a) Monomial

51. The value of $x^2 - 5$ at $x = -1$ is-

- (a) -2 (b) -1 (c) -4 (d) -5

Ans: (c) -4

52. $a^2 - b^2$ is a product of

- (a) $(a + b)(a - b)$ (b) $(a + b)(a + b)$ (c) $(a - b)(a - b)$ (d) None of these

Ans: (a) $(a + b)(a - b)$

53. Which of the following is the value of $(x + 1/x)^2$?

- (a) $x^2 + 1/x^2$ (b) $x^2 - 1/x^2$
(c) $x^2 + 1/x^2 + 2$ (d) $x^2 + 1/x^2 + 2x$

Ans: (c) $x^2 + 1/x^2 + 2$

54. Which of the following is obtained by subtracting $x^2 - y^2$ from $y^2 - x^2$?

- (a) $-2(x^2 - y^2)$ (b) $-2(x^2 + y^2)$ (c) $2(x^2 + y^2)$ (d) $2(x^2 - y^2)$

Ans: (a) $-2(x^2 - y^2)$

55. What degree does $x^3 - x^2y^2 - 8y^2 + 2$ have?

- (a) 2 (b) 3 (c) 4 (d) 7

Ans: (c) 4

56. What is the value of $5x^{25} - 3x^{32} + 2x^{-12}$ at $x=1$?

- (a) 0 (b) 2 (c) 4 (d) None of these

Ans: (c) 4

57. What is the product of $(x + a)$ and $(x + b)$?

- (a) $x^2 + (a - b)x + ab$ (b) $x^2 + (a + b)x - ab$
(c) $x^2 + (a + b)x - ab$ (d) $x^2 + (a + b)x + ab$

Ans: (a) $x^2 + (a - b)x + ab$

58. Which of the following is not a linear equation in one variable?

- (a) $33z + 5 = 0$ (b) $33(x + y) = 0$ (c) $33x + 5 = 0$ (d) $33y + 5 = 0$

Ans: (b) $33(x + y) = 0$

59. The solution of $2x - 3 = 7$ is:

- (a) 5 (b) 7 (c) 12 (d) 11

Ans: (a) 5

60. The solution of $2y + 9 = 4$ is:

- (a) $9/2$ (b) $4/9$ (c) $-2/5$ (d) $-5/2$

Ans: (d) $-5/2$

61. The solution of $y/5 = 10$ is:

- (a) 15 (b) 10 (c) 50 (d) 5

Ans: (c) 50

62. What should be added to $-7/3$ to get $3/7$?

- (a) $21/58$ (b) $58/21$ (c) $47/21$ (d) $50/21$

Ans: (b) $58/21$

Let the number be $x \Rightarrow -7/3 + x = 3/7 \Rightarrow x = 3/7 + 7/3 = (9+49)/21 = 58/21$

63. The age of the father is three times the age of the son. If the age of the son is 15 years old, then the age of the father is:

- (a) 50 years (b) 55 years (c) 40 years (d) 45 years

Ans: (d) 45 years

Let the age of the father is $x \Rightarrow x = 3 \times (\text{age of son}) = 3 \times (15) = 45 \text{ years}$

64. The difference between two whole numbers is 66. The ratio of the two numbers is 2: 5. The two numbers are:

- (a) 60 and 6 (b) 100 and 33 (c) 110 and 44 (d) 99 and 33

Ans: (c) 110 and 44

Let the two numbers be $2x$ and $5x$ since they are in the ratio of 2:5.
 The difference between $5x$ and $2x = 66 \Rightarrow 5x - 2x = 66 \Rightarrow x = 22$

65. Three consecutive integers add up to 51. The integers are:

- (a) 16,17,18 (b) 15,16,17 (c) 17,18,19 (d) 18,19,20

Ans: (a) 16, 17, 18

Let the three consecutive integers be $x, x+1, x+2$
 $x+(x+1)+(x+2) = 51 \Rightarrow 3x+3 = 51 \Rightarrow x = 48/3 = 16$

66. The solution for $3m = 5m - (8/5)$ is:

- (a) $8/5$ (b) $4/5$ (c) $5/4$ (d) $4/3$

Ans: (b) $4/5$

67. The degree of $x^2 - 5x + 2 = x^3$ is:

- (a) 3 (b) 2 (c) 1 (d) 0

Ans: (a) 3

68. The degree of equation $x^2 - 9 = 2x^2$ is:

- (a) 0 (b) 1 (c) 2 (d) 3

Ans: (c) 2

69. What is the value of x if $x + 9 = 12$?

- (a) 2 (b) 3 (c) 8 (d) 6

Ans: (b) 3

70. If a number is divided by 8 it gives 6 as the value. Find the number.

- (a) 36 (b) 42 (c) 48 (d) 56

Ans: (c) 48

Let X be the number $\Rightarrow X/8 = 6 \Rightarrow X = 8 \times 6 = 48$

71. Solve $2x + 9 = 4$.

- (a) $X = 6$ (b) $X = -5/2$ (c) $X = -3/2$ (d) $X = -9/2$

Ans: (b) $X = -5/2$

72. Find the value of x if $2x + 10 = 76$.

- (a) 33 (b) 36 (c) 66 (d) 32

Ans: (a) 33

73. The perimeter of a rectangle is 40 cm. If its width is 10 cm, then find the length.

- (a) 10 (b) 20 (c) 30 (d) 40

Ans: (a) 10

Perimeter of rectangle = $2(\text{length} + \text{width})$
 $40 = 2(x + 10) \Rightarrow 20 = x + 10 \Rightarrow x = 20 - 10 = 10$

74. If x is an even number, then the next even number is:
(a) $x+1$ (b) $x+2$ (c) $x+3$ (d) $x+4$

Ans: (b) $x+2$

75. The difference between the two numbers is 30. If the bigger number is x , then what is the smaller number?

(a) $x - 30$ (b) $30 - x$ (c) $30x$ (d) None of these

Ans: (a) $x - 30$

76. When a number is added to itself, it becomes 24. What is the number?

(a) 2 (b) 4 (c) 12 (d) 21

Ans: (c) 12

77. A polynomial contains _____ number of terms:

(a) One (b) Two (c) Three (d) Any

Ans: (d) Any

78. In which of the following, the two expressions are like terms?

(a) $7x$ and $7y$ (b) $7x$ and $9x$ (c) $7x$ and $7x^2$ (d) $7x$ and $7xy$

Ans: (b) $7x$ and $9x$

79. If we add, $7xy + 5yz - 3zx$, $4yz + 9zx - 4y$ and $-3xz + 5x - 2xy$, then the answer is:

(a) $5xy + 9yz + 3zx + 5x - 4y$ (b) $5xy - 9yz + 3zx - 5x - 4y$
(c) $5xy + 10yz + 3zx + 15x - 4y$ (d) $5xy + 10yz + 3zx + 5x - 6y$

Ans: (a) $5xy + 9yz + 3zx + 5x - 4y$

80. Which point lies only on y -axis?

(a) $(-2, 0)$ (b) $(2, 0)$ (c) $(0, -2)$ (d) $(2, -2)$

Ans: (c) $(0, -2)$

81. If we join $(-3, 2)$, $(-3, -3)$ and $(-3, 4)$, then we obtain:

(a) A triangle
(b) Straight-line passing through origin
(c) Straight-line without passing through origin
(d) None of the above

Ans: (c) Straight-line without passing through origin

82. The point $(4, 0)$ lies on which of the following?

(a) x - axis (b) y - axis (c) Origin (d) None of the above

Ans: (a) x - axis

83. The point $(-2, -2)$ is:

(a) Near to x - axis (b) Near to y - axis
(c) Near to origin (d) Equidistant from x - axis and y - axis.

Ans: (d) Equidistant from x - axis and y - axis

84. The point $(-2, 5)$ is nearer to:

(a) x - axis (b) y - axis (c) Origin (d) None of the above

Ans: (b) y - axis

85. The point $(-5, 2)$ is nearer to:
(a) x - axis (b) y - axis (c) Origin (d) None of the above
Ans: (a) x - axis

86. The point $(0, 0)$ lies at:
(a) x - axis B. y - axis (c) Origin (d) None of the above
Ans: (c) Origin

87. If we add, $7xy + 5yz - 3zx$, $4yz + 9zx - 4y$ and $-3xz + 5x - 2xy$, then the answer is:
(a) $5xy + 9yz + 3zx + 5x - 4y$. (b) $5xy - 9yz + 3zx - 5x - 4y$
(d) $5xy + 10yz + 3zx + 15x - 4y$ (d) $5xy + 10yz + 3zx + 5x - 6y$
Ans: (a) $5xy + 9yz + 3zx + 5x - 4y$

88. If we subtract $4a - 7ab + 3b + 12$ from $12a - 9ab + 5b - 3$, then the answer is:
(a) $8a + 2ab + 2b + 15$ (b) $8a + 2ab + 2b - 15$
(c) $8a - 2ab + 2b - 15$ (d) $8a - 2ab - 2b - 15$
Ans: (c) $8a - 2ab + 2b - 15$

89. If we multiply $5x$ and $(-4xyz)$, then we get:
(a) $20x^2yz$ (b) $-20x^2yz$ (c) x^2yz (d) $-2xyz$
Ans: (b) $-20x^2yz$

90. The product of $4x$ and 0 is:
(a) $4x$ (b) 4 (c) 0 (d) None of the above
Ans: (c) 0

91. The volume of a cuboid with length, breadth and height as $5x$, $3x^2$ and $7x^4$ respectively is:
(a) $105x^7$ (b) $105x^2$ (c) $105x^4$ (d) $105x$
Ans: (a) $105x^7$ cubic units

92. The product of $5x$ and $3y$ is:
(a) $x y$ (b) $2xy$ (c) $5xy$ (d) $15xy$
Ans: (d) $15xy$

93. The product of $6x$ and $-11x$ is:
(a) $66x^2$ (b) $-66x^2$ (c) x^2 (d) $-x^2$
Ans: $-66x^2$

94. The area of a rectangle whose length and breadth are $3y$ and $9y^2$ respectively is:
(a) $12y^3$ (b) $21y^3$ (c) $27y^3$ (d) y^3
Ans: (c) $27y^3$

95. The area of a rectangle that has length $= 2a^2b$ and breadth $= 3ab^2$ is:
(a) $6a^3b^3$ (b) a^3b^3 (c) $2a^3b^3$ (d) $4a^3b^3$
Ans: (a) $6a^3b^3$

96. The side of a cube is $2a$. Find the volume of the cube.

- (a) $4a^2$ (b) $2a$ (c) $8a^3$ (d) 8

Ans: (c) $8a^3$

97. Multiplication of monomials x^2 , $(-x)^3$, $(-x)^4$ is equal to:

- (a) x^9 (b) x^5 (c) x^7 (d) x^6

Ans: (a) x^9

98. The value of $(x - y)(x + y) + (y - z)(y + z) + (z - x)(z + x)$ is:

- (a) $x + y + z$ (b) $x^2 + y^2 + z^2$ (c) $xy + yz + zx$ (d) 0

Ans: (d) 0

99. $(a - b)^2$ is equal to:

- (a) $a^2 + b^2 - 2ab$ (b) $a^2 + b^2 + 2ab$ (c) $a^2 + b^2$ (d) $2ab$

Ans: (a) $a^2 + b^2 - 2ab$

100. The product of $3xy^2z$ and $4x$ is:

- (a) $12xyz$ (b) $12xy^2$ (c) $12x^2y^2z$ (d) $12x^2yz$

Ans: (c) $12x^2y^2z$